

Widad Dabbas

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Education

Durham University

MSc in Mathematical Sciences

Durham, UK

September 2024 - Present

- **Proposed Thesis:** Soil microbiome complexity-stability theory supported by RMT and novel metagenomic data methods

Ecoversity

Permaculture Design Certification

Remote, Punta Mona, Costa Rica

October 2023 - May 2024

- **Design Project Leader**

University of St Andrews

BSc in Biology and Mathematics with Honours

St Andrews, UK

Sept 2019 - June 2023

- **Biology Dissertation:** Disturbance and Niche Construction as Mechanisms of Community Assembly
- **Mathematics Dissertation:** Distinguishing knots with category theory

Amman Bacalaureate School

International Bacalaureate Diploma Programme

Amman, Jordan

August 2016 - June 2018

Experience

Soul Curated Wellness

Regenerative Design Consultant

Amman, Jordan

January - August 2024

- Initiated regenerative and permaculture design within the company.
- Conducted thorough site analyses using local historical data and on-site field surveys.
- Consulted on design queries, delivering research and solutions of regenerative materials and processes.
- Examined local ecosystems for their current state and resources and to inform mutualistic design opportunities.
- Collaborated to design onsite water structures, natural buildings and agriculture.
- Lead a team of 4 to produce a comprehensive master-plan including presentation, realistic timeline and budget.
- **Technical Skills:** Architectural Diagramming, Proposal Preparation, Project Scoping, Site Analysis, 3D modeling

Danish-Swedish summer school on Topological Data Analysis and spatial statistics, Aalborg University

Summer School Student and Conference Attendee

Aalborg, Denmark

June 2023

- Gained theoretical and practical understanding of spatial statistics.
- Gained theoretical and practical understanding of topological data analysis.
- Presented on the use of time series topological data analysis as a predictive tool.
- Used python to conduct topological data analysis on a set of three financial time series.
- **Technical Skills:** Python (Giotto-TDA, Scikit-TDA), Spatial Statistics, Topological Data Analysis

Control Theory and Systems Biology Lab, ETH Zurich

Visiting Student

Basel, Switzerland

August - September 2022

- Analysed discrete chemical reaction networks.
- Used MATLAB to simulate PIDs and integral controllers in biological networks.
- Assisted in molecular cloning to build theorised biological networks in yeast cells.
- Investigated the dynamical properties of simulated and constructed biological networks.
- **Technical Skills:** Control Systems Engineering, Dynamical Systems Analysis, Matlab, Molecular Cloning, Synthetic Biology, Numerical Optimisation

St Andrews University Kendo Club

Safety Officer

St Andrews, UK

May 2022 - June 2023

- Conducted a complete, up to date risk assessment.
- Maintained inventory of first aid kit including perishables.
- Improved safe training conditions, and provided first aid when necessary.
- Communicated with venues and foreign clubs in preparation for a local event.
- **Technical Skills:** Documentation, First aid, Risk Assessment

Royal Botanic Garden

Assistant Botanist

Balqa, Jordan

January - June 2022

- Practiced plant and seed identification techniques.
- Utilized irisBG GIS technology to map and analyze plant populations.
- Implemented various techniques to preserve samples for seed bank and herbarium.
- Conducted archival scanning and documentation of century old specimens using Herbscan.
- Gained expertise in seed and vegetative pretreatments, propagation and subsequent plant care methods.
- Gained insight into sediment and compost preparation, acknowledging its ecological and physical properties.
- **Technical Skills:** Archival Preservation, Field Sampling, GIS, IrisBG, Plant Care, Propagation, Sediment Preparation

Kalila wa Dimna Publishing

Amman, Jordan

Book Illustrator

July 2019 - August 2020

- Completed 10 unique watercolour illustrations.
- Captured plot, emotions and themes within watercolour illustration.
- Employed time management techniques to complete the illustrations alongside my studies.
- **Technical Skills:** Photoshop, Watercolour Illustration

Harvard Girguis Lab

Cambridge, MA, USA

Laboratory Intern

September - December 2018

- Temporarily joined the Girguis Lab for Ecophysiology, Biogeochemistry and Engineering.
- Independently constructed an OpenROV and used it to gather underwater footage in the Gulf of Aqaba.
- Collaborated on a video research proposal as a director, editor and translator, making use of ROV footage.
- Shadowed PhD research on *Sporomusa ovata*, and was responsible for independent work with anaerobic liquid cultures.
- **Technical Skills:** Anaerobic Microbiology, Electrical Wiring, Videography

Jackson Hole Science Media Awards at WGBH

Boston, MA, USA

Event Assistant

October 2018

- Assisted in running the Jackson Hole Science Media Awards and Summit at WGBH.
- Provided onsite event support, including catering, ushering and sound system support.
- Worked closely with the event management team to prepare the venue and delegate skill based tasks.
- Ran tours of virtual and augmented reality systems after learning the new technologies, their functioning and uses.

Ocean Exploration Trust, E/V Nautilus

Pacific Ocean, CA, USA

Science/Data Team and Video Engineer

July - August 2017

- Remotely operated the 23 live broadcast cameras on ROVs Hercules and ARGUS and aboard the E/V Nautilus.
- Coordinated live video streams to allow safe piloting of ROVs and surveillance by the scientific team.
- Engaged in live radio discussion and analysis of underwater footage with the science team.
- Prepared deep sea samples in the onboard wet lab for further use by onshore researchers.
- Controlled the zoom, focus, and iris of cameras to maintain high image quality.
- **Technical Skills:** Audio Engineering, Sample Preparation, Science Communication, Video Engineering

Learning Trail Nursery

Amman, Jordan

Summer School Teacher

June 2014 - June 2017

- Planned and led art lessons.
- Responsibly managed groups of children at 5-10 years of age.
- Took initiative and creativity in problem solving, conflict resolution and engaging teaching.
- Built strong relationships, and was asked to return with promotion over multiple years.

Projects

Earth's Whispers: Solo Exhibition

Amman, Jordan

Jordan National Gallery of Fine Arts

August 2024

- Successfully curated my debut solo exhibition as a first example of integrating science communication with fine arts in Jordan. Inspired by the wonders of soil as a vital ecosystem and an impressive natural material, I experimented with the dichotomy of modern glass prints supported by handmade adobe sculptures. Brief microbial and structural analysis was also conducted in collaboration with professors at the University of Jordan to produce a complimentary educational film.
- **Technical Skills:** Electron Microscopy, Photography, Science Communication, Soil Microbiology, Videography.

Knomhadique Permaculture Design

Burma, Jordan

Ecoversity

January - April 2024

- Completed the permaculture design for a wellness and community center as part of a 4 member team with the Ecoversity Permaculture Design Certification. Considered existing ancestral and scientific knowledge of the local ecosystem in its current state and at its most useful potential. Communicated between the client and team to identify the ideal use of site resources. Learned new technologies and building techniques to provide organic architectural designs and material suggestions. Applied client feedback and presented finalised design plan by collaborating to design onsite water structures, natural buildings and agriculture
- **Technical Skills:** Site Analysis, Architectural Diagramming, Topological Data Analysis.

Financial Time Series Topological Data Analysis (TDA)

Aalborg, Denmark

Aalborg University

June 2023

- Investigated and presented on the predictive capabilities of TDA inancial market conditions using financial time series data as part of the Danish-Swedish summer school on Topological Data Analysis and spatial statistics. Methods were adopted from the literature to analyse three financial time series of NVDA, WMT and XOM stock prices over the last 20 years. Interpreted TDA analyses with reference to financial and historical records whilst learning and using TDA libraries in Python. Presented findings in a conference setting. No conclusive results were made, but persistent entropy appeared to be negatively correlated with market volatility.
- **Technical Skills:** Python (giotto-tda,scikit-tda), Spatial Statistics, Time Series Analysis, Topological Data Analysis.

Experimental Research Project: Ecological Niche Construction

St Andrews, UK

University of St Andrews

September 2022 - April 2023

- Conducted research into the effects of ecological niche construction under various disturbance regimes under climate change and severe weather events as my biology dissertation. Research was primarily conducted by computational methods adapted from the literature. Links were then drawn between a defined ecological model and the existing knowledge and data from with coral reef ecosystems in mind.
- **Technical Skills:** Dynamical Systems Analysis, Ecological Modeling, MATLAB, R.

Project in Pure Mathematics: Knot invariants and Categorification

St Andrews, UK

University of St Andrews

September 2022 - April 2023

- Authored a dissertation walking through the use of categorification in knot theory. Independent study of category theory, homological algebra and knot theory was undertaken. Care was taken to break down and introduce complex topics to the reader through written or visual communication. A strong narrative was achieved and carried through technical definitions and reconstructed proofs.
- **Technical Skills:** InkScape, $\LaTeX$ (XY-pic), Technical Writing, Wolfram Mathematica.

Intertidal Project on *Ascophylum nodosum*

St Andrews, UK

University of St Andrews

August - September 2023

- Designed and conducted an observational study relating *A. nodosum* length to shoreline elevation. Line-intercept sampling was implemented to collect data. ANOVA and Akaike information criterion (AIC) were applied in R for data and statistical analyses. Results negatively correlated *A. nodosum* length with shoreline elevation irrespective of age. Limitations and explanations were discussed.
- **Technical Skills:** Line-Intercept Sampling, R, Statistical Data Analysis.

Red-Sea Trench Proposal

Cambridge, MA, USA

Harvard University

October 2018 - May 2019

- Collaborated on a video proposal to conduct research in the Gulf of Aqaba. Began by independently constructing an OpenROV during my time at the Girguis Lab for Ecophysiology, Biogeochemistry and Engineering. Gathered underwater footage in the Gulf of Aqaba by piloting the ROV off a boat. Directed, edited, translated and provided voiceover for the video research proposal.
- **Technical Skills:** Electrical Wiring, Proposal Writing, Videography

Skills

Programming	CSS, HTML, MATLAB, Python (Pandas,NumPy,Scikit-TDA), R (ggplot2,desolve), Wolfram Mathematica
Analytical	Experimental Design, Dynamical Systems Modeling/Analysis, Statistical Data Analysis, Time Series, Topological Data Analysis
Practical	Audio-Video Engineering, Electrical Wiring, Field Research, First Aid, Lab Skills, Microscopy, Species Identification, Synthetic Biology
Miscellaneous	Adobe, Blender, Diving (Free/SCUBA), Driving License, Illustration, $\LaTeX$ (Overleaf/R Markdown/XY-pic), Microsoft Office, Technical Writing
Soft Skills	Curiosity, Communication, Diagramming, Documentation, Engaging Presentation, Logic, Problem-solving, Teamwork
Languages	Arabic, English, French (conversational), Spanish (beginner)